

Grade 11 – Term 1

C3 Practice Activity 3 with Solutions

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Evaluated criterion

- C3: Applies probability distributions for periodic variables, calculates interval probabilities around a circle, and uses vector methods to describe a representative direction.

1 Introduction

A *circular distribution* is a probability distribution for a variable measured on a circle, such as an angle, a compass direction, or a time of day. It is not a density whose graph happens to be a circular arc. The defining feature is that the beginning and end of the scale are the same point: for example, 0° and 360° represent the same direction.

If Θ is measured in radians, then

$$\Theta \equiv \Theta + 2\pi.$$

Consequently a circular probability density function (pdf) is periodic:

$$f(\theta) = f(\theta + 2\pi), \quad \int_0^{2\pi} f(\theta) d\theta = 1. \quad (1)$$

As for an ordinary continuous distribution, probability is area under the density:

$$P(a \leq \Theta \leq b) = \int_a^b f(\theta) d\theta. \quad (2)$$

If an interval crosses 0, split it at the end of a chosen cycle. For example, on $[0, 2\pi)$,

$$P\left(\frac{11\pi}{6} \leq \Theta \leq \frac{\pi}{6}\right) = \int_{11\pi/6}^{2\pi} f(\theta) d\theta + \int_0^{\pi/6} f(\theta) d\theta. \quad (3)$$

This notation means the short arc travelled through 0, not the empty ordinary linear interval.

The simplest example is the *circular uniform distribution*:

$$f(\theta) = \frac{1}{2\pi}, \quad 0 \leq \theta < 2\pi. \quad (4)$$

Every direction is equally likely, so an arc of angular width L has probability $L/(2\pi)$. Thus

$$P\left(0 \leq \Theta \leq \frac{\pi}{2}\right) = \int_0^{\pi/2} \frac{1}{2\pi} d\theta = \frac{1}{4}.$$

Ordinary averages can be misleading for circular data: the arithmetic mean of 1° and 359° is 180° , although both directions are close to 0° . Instead, represent θ_i by the unit vector $(\cos \theta_i, \sin \theta_i)$ and calculate

$$\bar{C} = \frac{1}{n} \sum_{i=1}^n \cos \theta_i, \quad \bar{S} = \frac{1}{n} \sum_{i=1}^n \sin \theta_i, \quad \bar{\theta} = \text{atan2}(\bar{S}, \bar{C}). \quad (5)$$

If atan2 returns a negative angle, add 2π (or 360°) to report it on the chosen cycle. The mean resultant length $R = \sqrt{\bar{C}^2 + \bar{S}^2}$ indicates concentration: values near 1 mean the observations point in similar directions, while values near 0 mean there is little or no single representative direction.

“Circular distribution” names a category, not one particular model. Its members include the circular uniform distribution, the von Mises distribution (often used for angles in a role analogous to the normal distribution), and wrapped distributions such as the wrapped normal. This activity concentrates on the circular uniform model and circular means.

2 Periodic Structure and Uniform Probabilities

Problems:

2.1. A quarter-circle and periodicity

2.2. An interval that crosses zero

2.1 A quarter-circle and periodicity

Suppose Θ is uniformly distributed on $[0, 2\pi)$. **Questions/Tasks.**

C3.1 Verify that the density in (4) has total probability 1.

C3.2 Find $P(\pi/3 \leq \Theta \leq 5\pi/6)$ and explain why adding 2π to both endpoints does not change it.

C3.1 - Verify normalization.

The density is nonnegative and

$$\int_0^{2\pi} \frac{1}{2\pi} d\theta = \frac{2\pi}{2\pi} = 1.$$

It therefore satisfies both requirements for a pdf on one complete cycle.



C3.2 - Find and interpret the probability.

The angular width is $5\pi/6 - \pi/3 = \pi/2$. Therefore

$$P\left(\frac{\pi}{3} \leq \Theta \leq \frac{5\pi}{6}\right) = \frac{\pi/2}{2\pi} = \frac{1}{4}.$$

The endpoints $\pi/3 + 2\pi$ and $5\pi/6 + 2\pi$ describe the same directions one revolution later. Periodicity, $f(\theta) = f(\theta + 2\pi)$, makes the probability unchanged.



2.2 An interval that crosses zero

A wind direction Θ is uniformly distributed around the compass. Find the probability that it lies within 30° of north, where north is 0° . **Questions/Tasks.**

C3.1 Write the event as two intervals on $[0^\circ, 360^\circ)$.

C3.2 Calculate the probability and express it as a percentage.

C3.1 - Split the wrapped interval.

Directions within 30° of north run from 330° to 360° and from 0° to 30° . Thus the event is

$$\{330^\circ \leq \Theta < 360^\circ\} \cup \{0^\circ \leq \Theta \leq 30^\circ\}.$$

The two pieces are disjoint on the chosen representation of the cycle.



C3.2 - Calculate the probability.

The two arcs have total width $30^\circ + 30^\circ = 60^\circ$. Hence

$$P = \frac{60^\circ}{360^\circ} = \frac{1}{6} \approx 0.1667.$$

There is therefore a probability of about 16.67%. Treating $330^\circ \leq \Theta \leq 30^\circ$ as an ordinary linear interval would miss the circular meaning of the event.



3 Circular Means

Problems:

3.1. Directions near the cycle boundary

3.2. A mean with weak concentration

3.1 Directions near the cycle boundary

Two delivery riders leave a depot in directions 10° and 350° . **Questions/Tasks.**

C3.1 Explain why the arithmetic mean is inappropriate.

C3.2 Use unit vectors to find the circular mean and resultant length.

C3.1 - Compare the directions.

The arithmetic mean is $(10^\circ + 350^\circ)/2 = 180^\circ$, which points south. Both actual directions are only 10° from north, so this answer is an artefact of cutting the circular scale at 0° .



C3.2 - Calculate the circular mean.

Using $350^\circ = -10^\circ$ on the circle,

$$\bar{C} = \frac{\cos 10^\circ + \cos(-10^\circ)}{2} = \cos 10^\circ, \quad \bar{S} = \frac{\sin 10^\circ + \sin(-10^\circ)}{2} = 0.$$

Therefore

$$\bar{\theta} = \text{atan2}(0, \cos 10^\circ) = 0^\circ, \quad R = \cos 10^\circ \approx 0.9848.$$

The circular mean correctly points north, and R close to 1 shows strong agreement between the two directions.



3.2 A mean with weak concentration

Customer arrivals are grouped by position on a 24-hour clock. Three observed times correspond to angles 0° , 120° , and 240° . **Questions/Tasks.**

C3.1 Calculate $\bar{C}$ and $\bar{S}$.

C3.2 Decide whether a single average direction is meaningful.

C3.1 - Average the vectors.

The three unit vectors are equally spaced, and

$$\bar{C} = \frac{1 - \frac{1}{2} - \frac{1}{2}}{3} = 0, \quad \bar{S} = \frac{0 + \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}}{3} = 0.$$

Thus the average vector is the zero vector and $R = 0$.



C3.2 - Interpret the result.

Because $\text{atan2}(0, 0)$ has no defined direction, there is no unique circular mean for these data. This is informative rather than an error: the observations are perfectly balanced around the circle, so a single representative time would be misleading.



4 Applied Circular Probability

Problems:

4.1. Online orders by time of day

4.2. Choosing an appropriate model

4.1 Online orders by time of day

Under a circular uniform model for time of day, let H be the time on a 24-hour clock. A shop manager studies orders from 22:30 through 01:30. **Questions/Tasks.**

C3.1 Represent the event without treating midnight as an endpoint of time.

C3.2 Find and interpret its probability.

C3.1 - Represent the event.

On $[0, 24)$ hours the event is

$$\{22.5 \leq H < 24\} \cup \{0 \leq H \leq 1.5\}.$$

Equivalently, converting time h to angle by $\theta = 2\pi h/24$, it is the wrapped arc from $15\pi/8$ through 0 to $\pi/8$.



C3.2 - Find and interpret the probability.

The event lasts $1.5 + 1.5 = 3$ hours. Uniformity gives

$$P(22:30 \text{ through } 01:30) = \frac{3}{24} = \frac{1}{8} = 0.125.$$

The model assigns 12.5% of orders to that time window. This result depends on the uniform-model assumption; real order times may be concentrated at particular hours.



4.2 Choosing an appropriate model

An analyst considers three variables: daily revenue in pesos, wind direction, and the time of day at which peak electricity demand occurs. **Questions/Tasks.**

C3.1 Identify which variables are naturally circular and justify the choice.

C3.2 Name possible circular models when directions are uniform and when they cluster around one direction.

C3.1 - Identify circular variables.

Wind direction and peak-demand time are circular: $0^\circ = 360^\circ$, and midnight is both the end of one day and the beginning of the next. Revenue is linear; its largest and smallest values are not identified with each other. The shape of a graph alone does not determine whether a variable is circular.



C3.2 - Choose model families.

The circular uniform distribution is suitable if every direction is equally likely. If observations cluster around a preferred direction, a von Mises model may be suitable; a wrapped normal is another possible model. Data and context are needed to choose among them: “circular distribution” does not refer to one universal density.

